

FRACTIONS & OPERATIONS – LEVEL 1: EASY

Exercise 1

Evaluate and simplify fully:

$$\frac{5}{6} - \frac{3}{4} + \frac{2}{3}$$

Answer 1

1. Common denominator: $\text{LCM}(6, 4, 3) = 12$

2. Convert terms:

$$= \frac{10}{12} - \frac{9}{12} + \frac{8}{12}$$

3. Add numerators:

$$= \frac{9}{12} = \frac{3}{4}$$

Exercise 2

Evaluate and express in simplest form:

$$\left(\frac{4}{9} \times \frac{15}{8} \right) \div \frac{5}{6}$$

Answer 2

1. Multiply first:

$$\frac{4 \cdot 15}{9 \cdot 8} = \frac{60}{72} = \frac{5}{6}$$

2. Divide by second fraction:

$$= \frac{5}{6} \div \frac{5}{6} = 1$$

$$= 1$$

Exercise 3

Simplify the basic nested fraction:

$$\frac{1 - \frac{1}{3}}{2 + \frac{1}{2}}$$

Answer 3

1. Numerator: $1 - \frac{1}{3} = \frac{2}{3}$

2. Denominator: $2 + \frac{1}{2} = \frac{5}{2}$

3. Outer/Inner rule:

$$= \frac{2/3}{5/2} = \frac{2 \cdot 2}{3 \cdot 5}$$

$$= \frac{4}{15}$$

FRACTIONS & OPERATIONS – LEVEL 2: MEDIUM

Exercise 4

Evaluate mixed operations with fractions:

$$\frac{3}{4} \div \left(\frac{2}{5} + \frac{1}{2} \right) \times \frac{3}{5}$$

Answer 4

1. Parentheses: $\frac{2}{5} + \frac{1}{2} = \frac{4+5}{10} = \frac{9}{10}$
2. Divide: $\frac{3}{4} \div \frac{9}{10} = \frac{3}{4} \times \frac{10}{9} = \frac{5}{6}$
3. Multiply: $\frac{5}{6} \times \frac{3}{5}$

$$= \frac{1}{2}$$

Exercise 5

Simplify the nested fraction:

$$2 - \frac{1}{1 + \frac{1}{3}}$$

Answer 5

1. Bottom denominator: $1 + \frac{1}{3} = \frac{4}{3}$
2. Reciprocal fraction: $\frac{1}{4/3} = \frac{3}{4}$
3. Outer subtraction: $2 - \frac{3}{4}$

$$= \frac{5}{4}$$

Exercise 6

Simplify the complex fractional quotient:

$$\frac{\frac{3}{4} - \frac{1}{6}}{\frac{2}{3} + \frac{1}{2}}$$

Answer 6

1. Numerator: $\frac{9-2}{12} = \frac{7}{12}$
2. Denominator: $\frac{4+3}{6} = \frac{7}{6}$
3. Simplify quotient:

$$= \frac{7/12}{7/6} = \frac{7 \cdot 6}{12 \cdot 7}$$

$$= \frac{1}{2}$$

Exercise 7

Simplify the algebraic fraction:

$$\frac{1 - \frac{1}{x^2}}{1 + \frac{1}{x}}$$

Answer 7

1. Combine top & bottom:

$$\text{Top} = \frac{x^2 - 1}{x^2}, \quad \text{Bottom} = \frac{x + 1}{x}$$

2. Factor top numerator: $(x - 1)(x + 1)$

3. Simplify: $\frac{(x-1)(x+1)}{x^2} \cdot \frac{x}{x+1}$

$$= \frac{x-1}{x}$$

FRACTIONS & OPERATIONS – LEVEL 3: HARD

Exercise 8

Simplify the deeply nested continued fraction:

$$1 + \frac{1}{2 + \frac{1}{3 + \frac{1}{4}}}$$

Answer 8

1. Innermost denominator: $3 + \frac{1}{4} = \frac{13}{4}$
2. Invert: $\frac{1}{13/4} = \frac{4}{13}$
3. Next level: $2 + \frac{4}{13} = \frac{30}{13}$
4. Invert: $\frac{1}{30/13} = \frac{13}{30}$
5. Final sum: $1 + \frac{13}{30}$

$$= \frac{43}{30}$$

Exercise 9

Evaluate the mixed nested operation completely:

$$\frac{\frac{1}{2} - \frac{1}{3} \times (1 - \frac{1}{4})}{2 \div (1 + \frac{1}{2}) - \frac{5}{6}}$$

Answer 9

1. Numerator: $1 - \frac{1}{4} = \frac{3}{4} \Rightarrow \frac{1}{3} \cdot \frac{3}{4} = \frac{1}{4}$

$$\text{Top} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

2. Denominator: $1 + \frac{1}{2} = \frac{3}{2} \Rightarrow 2 \div \frac{3}{2} = \frac{4}{3}$

$$\text{Bottom} = \frac{4}{3} - \frac{5}{6} = \frac{8-5}{6} = \frac{1}{2}$$

3. Final quotient: $\frac{1/4}{1/2}$

$$= \frac{1}{2}$$

Exercise 10

Simplify the nested algebraic continued fraction:

$$\frac{x}{1 - \frac{x}{1 + \frac{x}{1-x}}}$$

Answer 10

1. Innermost denominator: $1 + \frac{x}{1-x} = \frac{1}{1-x}$
2. Fraction above: $\frac{x}{1/(1-x)} = x - x^2$
3. Main denominator: $1 - (x - x^2) = 1 - x + x^2$
4. Full expression:

$$= \frac{x}{1-x+x^2}$$

Exercise 11

Simplify the complex algebraic fraction fully:

$$\frac{\frac{1}{x+h} - \frac{1}{x}}{h} \div \left(1 + \frac{h}{x}\right)$$

Answer 11

1. Numerator bracket: $\frac{x-(x+h)}{x(x+h)} = \frac{-h}{x(x+h)}$
2. Divide by h : $\frac{-1}{x(x+h)}$
3. Second term: $1 + \frac{h}{x} = \frac{x+h}{x}$
4. Divide: $\frac{-1}{x(x+h)} \cdot \frac{x}{x+h}$

$$= -\frac{1}{(x+h)^2}$$